Particle Physics Homework 6

1). Show that the Green $f^{\underline{n}}$:

$$G(x^\mu - y^\mu) = \int \frac{d^4 p}{(2\pi)^4} \frac{i}{p_\mu p^\mu} e^{-i p_\mu (x^\mu - y^\mu)}$$

corresponds to the equation:

$$\partial_\mu \partial^\mu \phi(x^\mu) = 0$$

2). Calculate the Feynman propagator

$$\Delta_F(x^\mu - y^\mu) = \int \frac{d^3\vec{p}}{(2\pi)^3} \int dp^0 \frac{i}{(p^0)^2 - (\vec{p})^2 - m^2 + i\varepsilon} e^{-i p_\mu (x^\mu - y^\mu)}$$

for $y^0 > x^0$ using the Upper Half Plane (UHP)

3). Discuss what would happen if we write

$$\Delta'_F(x^\mu - y^\mu) = \int \frac{d^3\vec{p}}{(2\pi)^3} \int dp^0 \frac{i}{(p^0)^2 - (\vec{p})^2 - m^2 - i\varepsilon} e^{-i p_\mu (x^\mu - y^\mu)}$$

instead?

4). Find the mass dimension of the field & parameters in the Lagrangian

4.1) $\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 - m A^\mu \phi \partial_\mu \phi - \lambda^\mu \partial_\mu \phi$

4.2) $\mathcal{L} = \frac{1}{2} \gamma^\mu \psi \partial_\mu \psi - \frac{1}{2} m \psi \psi - \lambda \psi \psi \psi \psi$

$[\gamma^\mu] = 0$